DHANYASRI P K

Analyst | Generative AI | Backend Development

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ABOUT ME

AI Platform Engineer with hands-on experience in Generative AI, LLM orchestration, LangChain, LangGraph, MCP, RAG pipelines, and AWS Bedrock. Skilled in building AI-powered backend systems using FastAPI, vector databases, and scalable cloud-based workflows. Passionate about developing intelligent, production-oriented AI applications and backend architectures.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering
Bannari Amman Institute of Technology, Sathyamangalam 8.15 CGPA
Nov 2022- Apr 2026

HSC (Matric Board)
SRV HI-Tech Higher secondary School,Rasipuram.
Secured 94%
June 2021 - March 2022

SSLC (Matric Board)
SRV HI-Tech Higher secondary School,Rasipuram.
Secured 87%
June 2019 - March 2020

EXPERIENCE

ANALYST — AVASOFT Pvt Ltd

Dec 2025 – Aug 2026 | Chennai, Tamil Nadu

- Contributed to enterprise Generative AI and Agentic AI applications using Python, Strands Agent Framework, LangChain, LangGraph, MCP, RAG pipelines, and AWS Bedrock.
- Developed and integrated AI-powered backend services and REST APIs using FastAPI, supporting LLM-driven workflows and enterprise application requirements.
- Worked on multi-agent architectures, tool calling, knowledge-base/RAG retrieval, authentication workflows, and enterprise system integrations.
- Worked with PostgreSQL, vector databases, embeddings, and semantic search for AI-powered information retrieval and context-aware responses.
- Gained hands-on experience with Docker, Git, AWS services, deployment workflows, and Kubernetes concepts in enterprise AI application development.

PROJECTS

UATP Multi-Agent AI Support Platform

- Worked on an enterprise AI-powered multi-agent support platform built using the Strands Agent Framework, supporting UATP airline issuers including Air India, Air Niugini, and LATAM Airlines Chile.
- Worked with five specialized agents — Reports, SFTP, API, DataPull, and Statements — designed to handle issuer-specific operational and information requests.
- Integrated specialized tools including issuer lookup, API client lookup, IIN validation, template lookup, and knowledge-base/RAG search to provide contextual issuer-specific responses.
- Integrated operational workflows with Jira and email automation, enabling automated ticket creation and communication with the UATP member technical team for requests such as report, statement, and setup-related issues.
- Worked on issuer validation workflows using IIN, JWT, and domain-based authentication before allowing issuer-specific operations.

Bean & Brew Cafe Chatbot — LangChain, LangGraph & MCP

- Built a conversational AI café assistant using LangChain, LangGraph, and MCP, with tools for menu, ordering, and order-status workflows.
- Implemented PostgreSQL-backed conversation checkpointing to maintain workflow state and conversational context.

AI Coder Agent — LangGraph

- Developed an agentic coding workflow using LangGraph for coding-task analysis, code generation, and workflow orchestration.
- Integrated tool-based execution and state management to support iterative coding workflows.

Medical Document Intelligence — AWS Bedrock Data Automation

- Built an AI-powered document processing workflow to extract structured information such as diagnoses and prescription details from medical PDFs.

TECHNICAL SKILLS

Programming Languages: Java,Python

Frontend Technologies: HTML, CSS

Backend Technologies: FastAPI,REST API

Generative AI & Agentic AI: LangChain, LangGraph, MCP, FastMCP, RAG, LLM Orchestration, Multi-Agent Systems, Tool Calling, HITL, Embeddings, Semantic Search

Databases & Vector Search: PostgreSQL, pgvector, MySQL, Vector Databases

Cloud & AWS Services: AWS Bedrock, Amazon Nova, Bedrock Data Automation, Amazon S3,Amazon SES

DevOps & Infrastructure: Docker, Kubernetes, Git, Jira